

the dynamic nature of cloud environments enables changes to be made by anyone, from anywhere, through various means of interaction with the cloud.

In this landscape, dynamic provisioning is performed by individuals with varying levels of understanding. Hence, any misconfigurations or misprovisioning are easily exposed to the cloud provider, potentially leading to cost leakage and security risks when exposed to the internet. This amplifies the security and cost challenges compared to traditional on-premises setups. Regrettably, many enterprises face increased expenses or compromised security postures in the cloud compared to on-premises environments due to these reasons.

The challenges

Let's explore why cloud security and cost optimisation are closely intertwined challenges.

Design flaws

Both security and cost optimisation are fundamental aspects that should be deeply rooted in the design, architecture, development practices, and implementation of the cloud environment. When the architecture is not optimised for cost efficiency and security, it can result in unnecessary resource utilisation, inefficient workflows, misconfigurations or overprovisioning of cloud resources. Enterprises often treat security and cost as afterthoughts.

Complexity and obscurity

The dynamic nature of the cloud, with its vast array of services, configurations, and data, can create a lack of visibility and understanding. The obscurity inherent in the cloud environment can contribute to misconfigurations that can expose sensitive data or lead to unauthorised access.

Inefficient resource allocation is another consequence of the lack of understanding and visibility. Without clear insights into resource utilisation and performance, enterprises may inadvertently provision resources with higher capacity than necessary, leading to cost leakage.

Limited control

Enterprises often rely on distributed teams to handle security implementation and cost optimisation. However, limited control over these decentralised teams can hamper the effectiveness of security monitoring and cost optimisation efforts.

In an enterprise cloud landscape, ensuring coordination across teams, implementing unified controls and enforcing best practices becomes difficult, leading to inconsistencies in security practices and cost optimisation strategies.

Varying business objectives

Enterprises and business units often have diverse business objectives when it comes to cloud usage. Some prioritise agility, while others prioritise cost optimisation or compliance. However, aligning these divergent priorities and goals can be challenging, resulting in fragmented security practices and inefficient cost management.

This misalignment can occur due to the varying needs, priorities, and perspectives of different stakeholders within the enterprise. This can lead to a lack of cohesive strategy, inconsistent implementation of security measures, and suboptimal cost management approaches.

Reactive approach

Enterprises often prioritise security and cost only when situations escalate beyond their control, such as a security breach or cost bleed. Addressing security and cost considerations reactively when

situations escalate beyond control can be extremely expensive.

Potholes

Enterprises often overlook the importance of being aware of potential cost leakage that can be exploited by cloud service providers due to misprovisioning or misconfigurations in their cloud implementations, similar to how attackers continuously exploit vulnerabilities in security systems.

Assumptions

Failing to validate the effectiveness of security controls or the capacity of provisioned resources can pose risks and negatively impact both security and cost management. When assumptions are made without proper benchmarking or validations, vulnerabilities may arise, potentially compromising the system's security.

Likewise, assuming resource capacity without adequate benchmarking can result in overprovisioned and underutilised resources, leading to unavoidable expenses or performance issues.

The right approaches

While there are similarities in the challenges, there are also similarities in the approaches to remediate them. Embracing this synergy empowers enterprises to not only conquer challenges but to architect a cloud environment that thrives on proactive enhancements.

Proactive approach

Integrating security and cost considerations early in the design and implementation stages is vital for a secure and cost-efficient cloud environment. By incorporating multilayered security controls and comprehensive cost controls from the start, enterprises can prevent vulnerabilities and inefficiencies. Proper benchmarking and capacity planning enable understanding of